Power BI Basics

Question 1: What is Power BI and why is it used in businesses?

Answer : Power BI is a business intelligence and data visualization platform developed by Microsoft that enables users to convert data from various sources into interactive dashboards, reports, and data-driven insights.

Why it is used in businesses:

**Informed Decision-Making: Helps organizations track key performance indicators (KPIs) and business metrics in real-time.

**Data Consolidation: Combines multiple disparate data sources (Excel, SQL databases, cloud services) into a single, unified view.

**Interactive Visualizations: Translates complex numerical data into intuitive charts, graphs, and maps.

**Collaboration: Facilitates secure sharing and collaboration across teams using the Power BI Service.

Question 2: Name and explain the three main components of Power BI.

Answer : The three main components of Power BI are:

1. Power BI Desktop: A free desktop authoring and development tool used to connect to data, clean/transform it, build data models, and create reports.
2. Power BI Service (SaaS):A cloud-based online service used for publishing, sharing, collaborating on reports, and creating dashboards.
3. Power BI Mobile: Mobile applications available for iOS and Android devices that allow users to view and interact with reports and dashboards on the go.

Question 3: Explain the Power BI workflow.

Answer : The standard workflow in Power BI follows these steps:

1. Connect: Import data from various sources (files, databases, web services).
2. Transform (Power Query): Clean, shape, and format the data (handling missing values, changing data types, merging tables).
3. Model: Establish relationships between tables, create calculated columns, and write measures using DAX (Data Analysis Expressions).
4. Visualize: Build visual reports containing charts, matrices, maps, and slicers.
5. Publish & Share: Publish the reports to the Power BI Service to share insights with stakeholders.

Question 4: List any four data cleaning tasks that can be performed in Power Query.

Answer : Four common data cleaning tasks in Power Query include:

1. Removing Duplicates or Nulls: Eliminating duplicate rows or filtering out missing values.
2. Changing Data Types: Converting text values to numerical, date/time, or boolean formats.
3. Splitting Columns: Separating a single column into multiple columns based on a delimiter (e.g., space or comma).
4. Trimming and Cleaning Text: Removing extra white spaces or non-printable characters from text fields.

Question 5: Write step by step instructions to load the above dataset (`Real_Estate Dataset`).

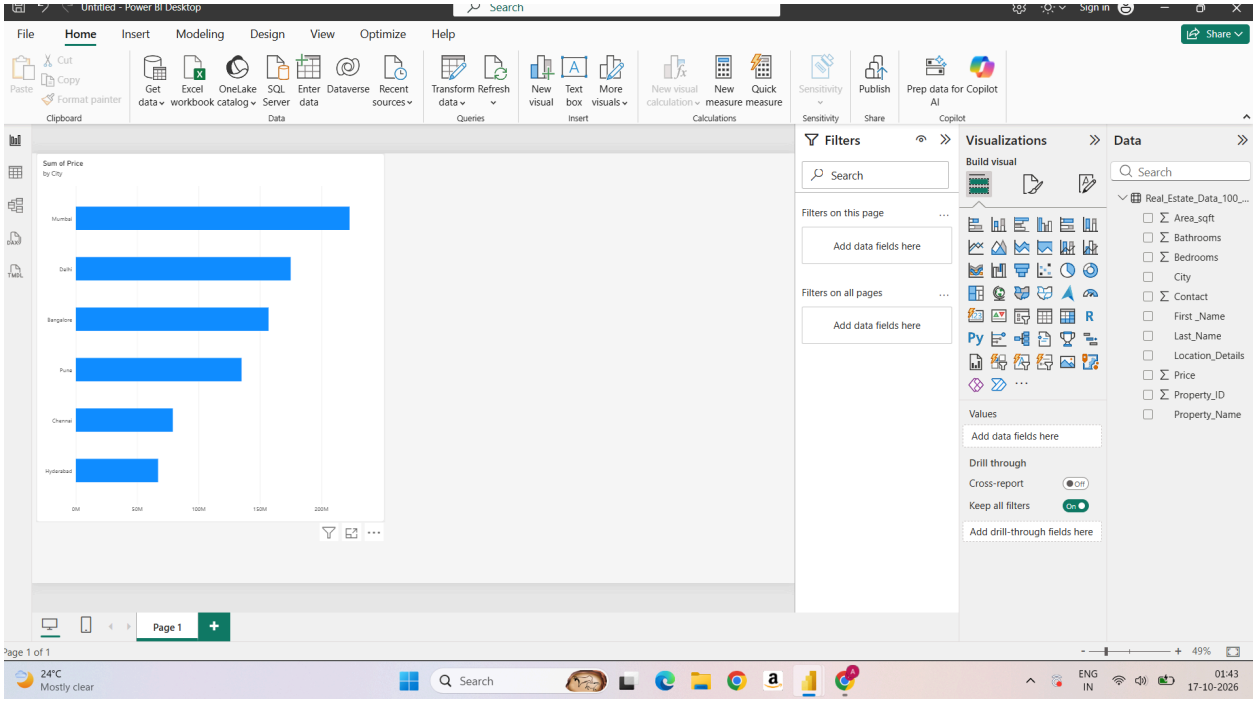
Answer : To load the dataset into Power BI Desktop:

1. Open Power BI Desktop.
2. On the Home ribbon, click on Get Data (select the appropriate file connector, such as Text/CSV or Excel workbook depending on the dataset format).
3. Browse to locate and select the `Real_Estate Dataset` file, then click Open.
4. In the Navigator window, check the box next to the table or sheet containing the dataset.
5. Click Transform Data if you need to clean the data in Power Query, or click Load to import the dataset directly into Power BI.

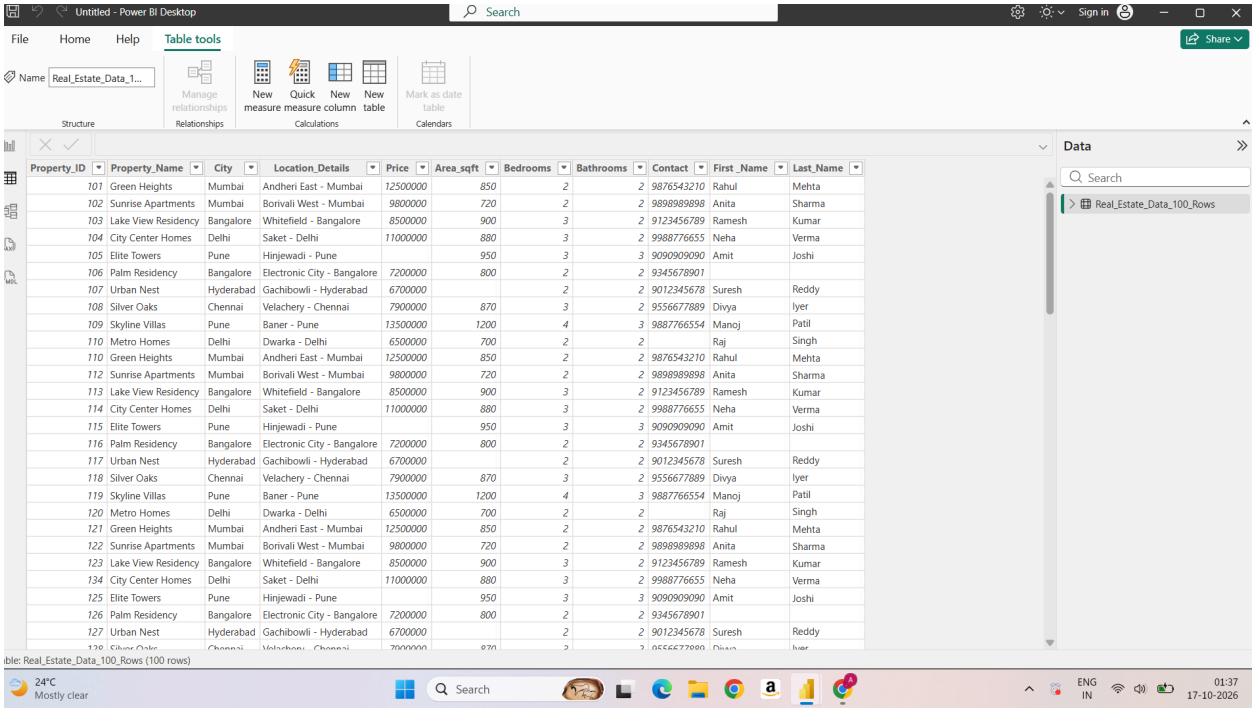
Question 6: Define Data View, Report View, and Model View. Explain the purpose of each view.

Answer : Power BI Desktop provides three distinct views accessed from the left-hand navigation pane:

1. Report View:
*Definition: The primary canvas where visualizations, charts, and report pages are designed.
*Purpose: To build and present visual stories and interactive dashboards for end-users.



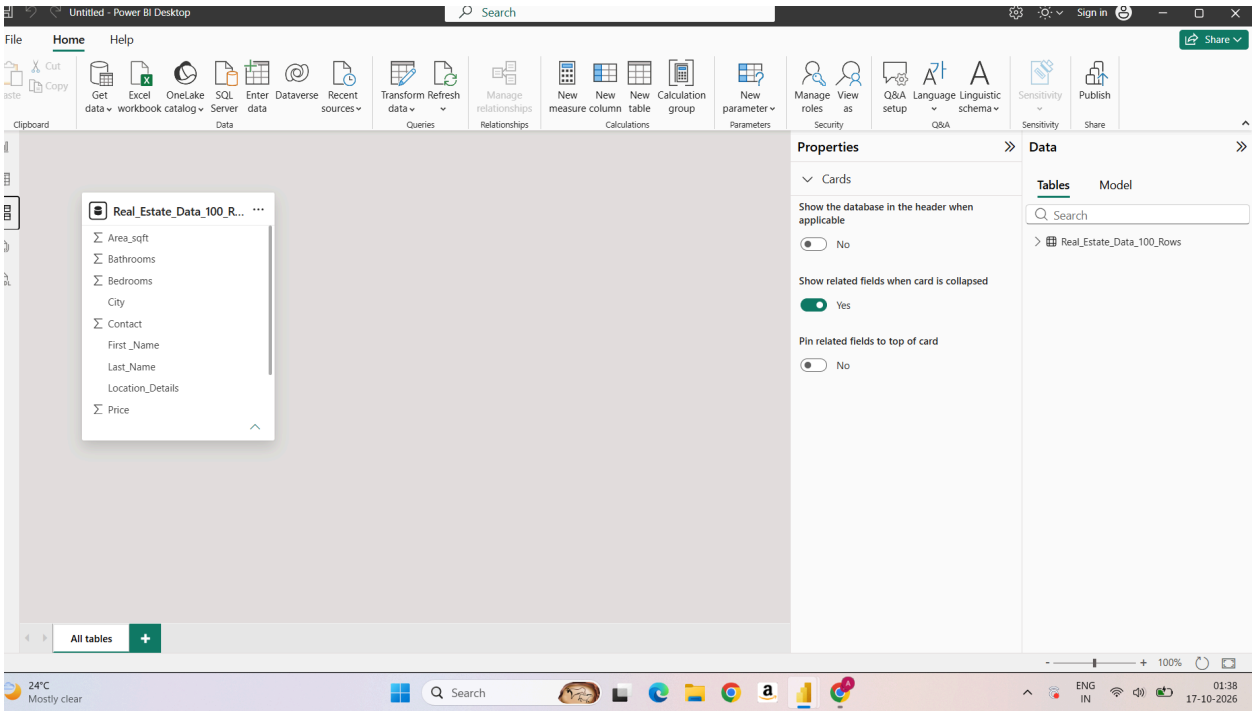
2. Data View:
*Definition: A tabular view displaying the raw data rows and columns loaded into the data model.
*Purpose: To inspect data values, create calculated columns, and review table structures.



3. Model View:

*Definition: A diagrammatic interface displaying all tables and their interconnecting relationships.

*Purpose: To manage table relationships, configure cross-filter directions, and organize complex data models.



Question 7: Discuss the different data sources that Power BI supports.

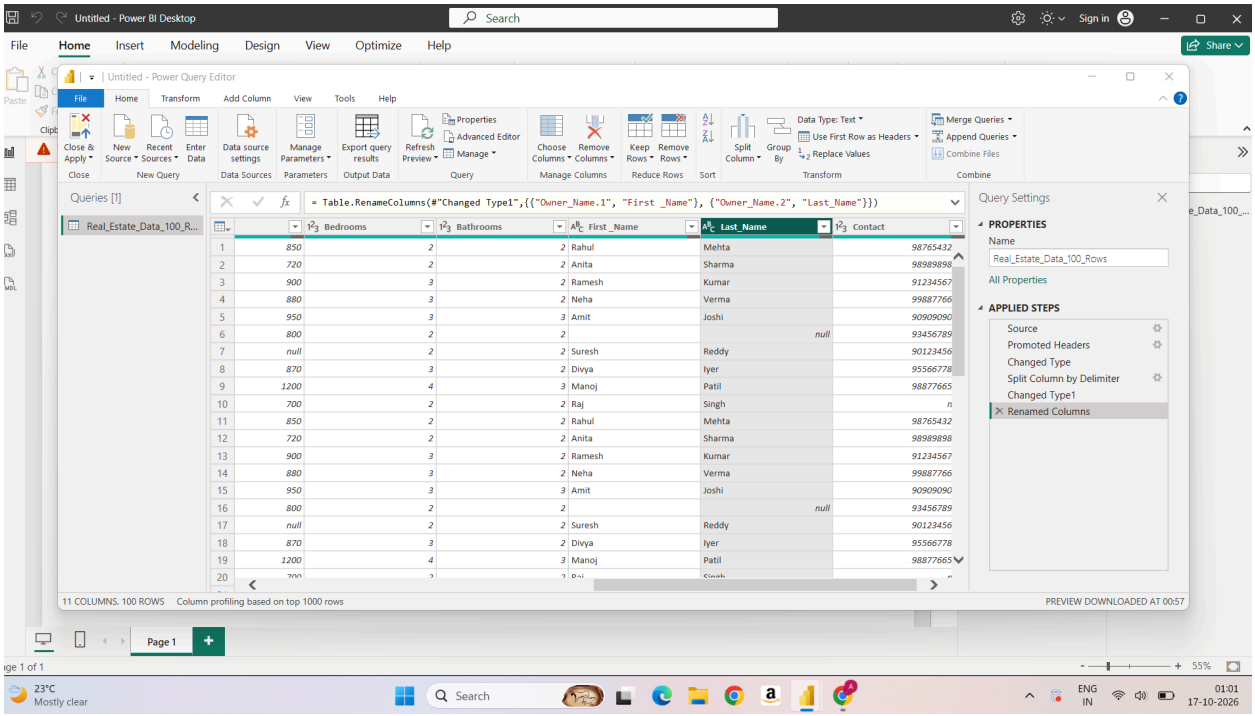
Answer : Power BI supports a wide array of data sources, categorized into:

* File Sources: Excel (.xlsx), CSV, XML, JSON, text files, and PDFs.
* Databases: SQL Server, Oracle, MySQL, PostgreSQL, IBM DB2, and Snowflake.
* Cloud & Online Services: Azure SQL Database, SharePoint lists, Google Analytics, Salesforce, and GitHub.
* Other Sources: Web pages (HTML scraping), OData feeds, ODBC/OLEDB connections, and blank queries.

Question 8: Split Owner Name to create two new columns as First Name and Last Name.

Answer : To split the "Owner Name" column using Power Query:

1. Open the Power Query Editor.
2. Select the Owner Name column.
3. Go to the Home or Transform tab on the ribbon and click Split Column -> By Delimiter.
4. Choose the appropriate delimiter (e.g., Space if the name is structured as "First Last") and select Left-most delimiter or Each occurrence.
5. Click OK. Power Query will split the column into two separate columns.
6. Double-click the newly generated column headers and rename them to First Name and Last Name.
7. Click & Apply on the Home tab to load the updated data back into Power BI.


Untitled - Power BI Desktop
Search
Sign in
File
Home
Insert
Modeling
Design
View
Optimize
Help
Share
Untitled - Power Query Editor
File
Home
Transform
Add Column
View
Tools
Help
Close & Apply
New Source
Recent Sources
Enter Data
Data source settings
Manage Parameters
Export query results
Refresh Preview
Properties
Advanced Editor
Manage
Choose Columns
Remove Columns
Keep Rows
Remove Rows
Split Column
Group By
Data Type: Text
Use First Row as Headers
Replace Values
Merge Queries
Append Queries
Combine Files
Close
New Query
Data Sources
Parameters
Output Data
Query
Manage Columns
Reduce Rows
Sort
Transform
Combine
Queries [1]
Real_Estate_Data_100_R...
= Table.RenameColumns(#"Changed Type1",{{"Owner_Name.1", "First _Name"}, {"Owner_Name.2", "Last_Name"}})
Bedrooms
Bathrooms
First _Name
Last_Name
Contact
1 850 2 2 Rahul Mehta 98765432
2 720 2 2 Anita Sharma 98989898
3 900 3 2 Ramesh Kumar 91234567
4 880 3 2 Neha Verma 99887766
5 950 3 3 Amit Joshi 90909090
6 800 2 2 null 93456789
7 null 2 2 Suresh Reddy 90123456
8 870 3 2 Divya Iyer 95566778
9 1200 4 3 Manoj Patil 98877665
10 700 2 2 Raj Singh n
11 850 2 2 Rahul Mehta 98765432
12 720 2 2 Anita Sharma 98989898
13 900 3 2 Ramesh Kumar 91234567
14 880 3 2 Neha Verma 99887766
15 950 3 3 Amit Joshi 90909090
16 800 2 2 null 93456789
17 null 2 2 Suresh Reddy 90123456
18 870 3 2 Divya Iyer 95566778
19 1200 4 3 Manoj Patil 98877665
Query Settings
PROPERTIES
Name
Real_Estate_Data_100_Rows
All Properties
APPLIED STEPS
Source
Promoted Headers
Changed Type
Split Column by Delimiter
Changed Type1
Renamed Columns
11 COLUMNS, 100 ROWS
Column profiling based on top 1000 rows
PREVIEW DOWNLOADED AT 00:57
Page 1
55%
23°C
Mostly clear
Search
ENG
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17-10-2026